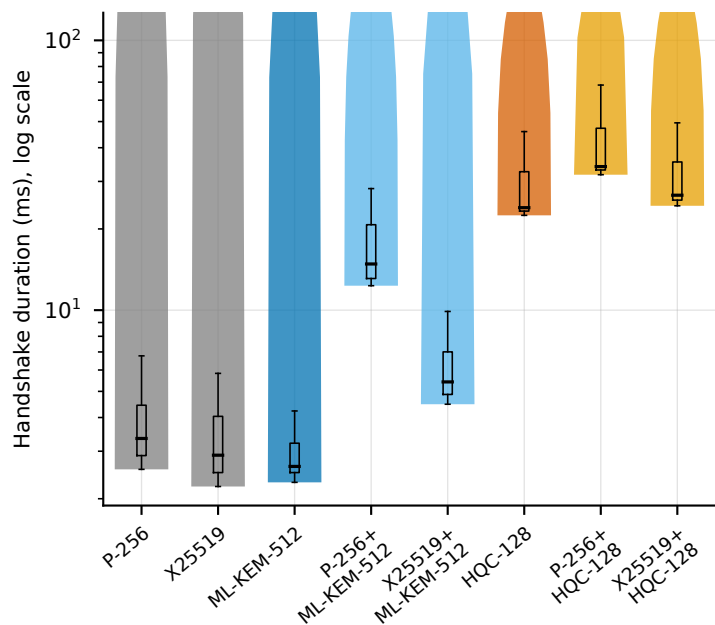


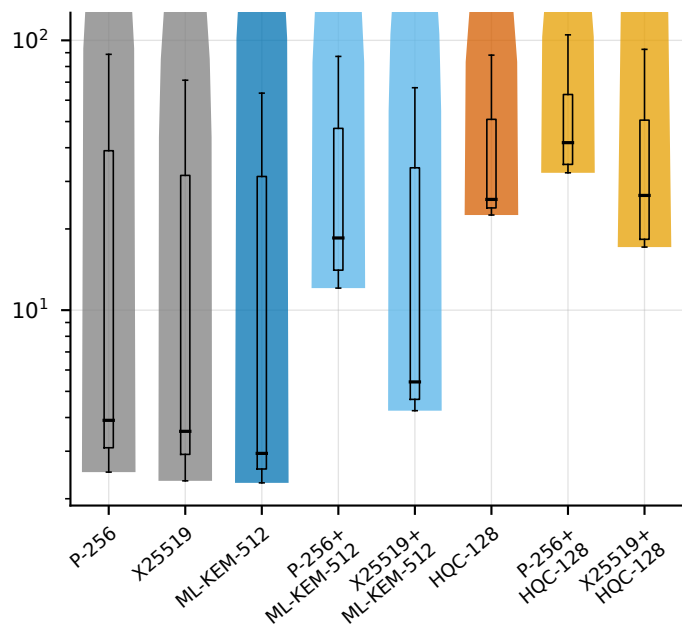
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	332.8	296.4	7.00k	2/500	351.6	245.6	9.00k	0
X25519	404.8	273.8	12.00k	5/500	239.7	250.4	4.10k	0
ML-KEM-512	271.2	272.9	7.00k	0	330.0	293.3	9.86k	0
P-256+ML-KEM-512	246.9	213.6	3.02k	0	286.4	258.3	7.02k	0
X25519+ML-KEM-512	237.4	241.9	7.00k	0	348.4	220.8	5.00k	0
HQC-128	109.5	333.4	3.13k	0	157.7	317.8	6.03k	0
P-256+HQC-128	104.1	375.1	7.02k	0	206.3	292.4	7.03k	0
X25519+HQC-128	101.7	342.3	3.05k	0	166.3	379.4	11.42k	1/500